

What Small Models Know About Kazakh Morphology and Cannot Be Asked

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Abstract

Multiple-choice benchmarks report that language models underperform in Kazakh relative to Russian, but cannot say why: a model can answer them without ever producing a Kazakh word form. We introduce a generative benchmark of Kazakh nominal inflection covering eight morphological categories over fifty regular noun stems, with gold answers computed from vowel-harmony and consonant-assimilation rules rather than annotated by hand. Holding items and gold answers byte-identical while varying only the language of the instruction, we find that small models are limited by instruction comprehension rather than by morphological knowledge. gemma3-4b scores 16.6% under Kazakh instructions against 48.8% in Russian and 52.0% in English, with a noise floor of 2.7 points; qwen2.5-3b scores 4.3% and 21.0%. The deficit is not uniform: categories whose Kazakh grammatical terminology is common show little or no penalty, while those with more specialised terminology fall to zero. A single in-context example raises Kazakh-instructed accuracy to 54.7% and five raise it to 74.7%, eliminating the gap, with gains concentrated precisely in the categories that had failed — the knowledge was present throughout and the instruction could not reach it. This reconciles our findings with prior work reporting a 1-5 point instruction-language penalty for Turkish and Finnish under few-shot prompting: demonstrations substitute for instruction comprehension and therefore mask the effect. The gap also vanishes with scale, gemma-4-31b reaching 98.7% and 100.0% with one genuine morphological error in 300 items. We further report five methodological findings for cross-lingual generative evaluation, including a translated case label that moved one category by 41 points and was fully rescued by examples, a morpheme that resists isolated elicitation on grammatical grounds, and hidden reasoning tokens that silently consume the output budget and are scored as errors.

1. Introduction

Kazakh is spoken by roughly fourteen million people and is an official language of a country of twenty million, yet remains among the languages for which large language models are least reliable. It is also agglutinative: nominal stems take suffixes whose surface form is determined jointly by vowel harmony, which propagates the front/back quality of the last stem vowel, and by consonant assimilation, which selects the suffix onset according to the voicing and manner of the stem-final segment. A model that has memorised frequent word forms and a model that has acquired the underlying rules are indistinguishable on any benchmark that does not require the model to *produce* a form it may never have seen.

Existing Kazakh benchmarks do not require this. KazMMLU (Togmanov et al., 2025), the largest and most widely used, comprises 23,000 multiple-choice questions; a model can score well on all of them without emitting a single Kazakh word form. Its authors observe that models perform better in Russian than in Kazakh and attribute the gap to "differences in training data availability, language complexity, or tokenization differences," without distinguishing between them. Multiple-choice evaluation cannot make that distinction: it conflates knowing something with being able to say it.

We introduce a generative benchmark for Kazakh nominal morphology in which gold answers are derived from phonological rules rather than annotated by hand, making the item set reproducible from a seed, cheap to extend, and free of annotator disagreement. Each item presents a citation-form noun and a target morphological category and requires the model to produce the inflected form. Because the gold answer is computed, the analysis can be conditioned on the linguistic properties that

determine it — harmony class, stem-final sound, and the specific allomorph — so that we report where a model fails rather than only how often.

Our central experiment holds items and gold answers byte-identical while varying only the language of the instruction. Small models prove limited by instruction comprehension rather than morphological knowledge. That English performs at least as well as Russian rules out an explanation grounded in Kazakhstan's bilingual situation: the effect is instruction comprehension in general, not transfer from a contact language. Underneath that parity, however, the choice of instruction language shifts individual categories by 20–30 points in opposite directions (Section 5.2). A single in-context example then recovers most of the deficit and five eliminate it, showing that the knowledge was present and the instruction could not reach it (Section 5.3). The morphology is largely present in models that appear, when addressed in Kazakh, not to have it.

2. Related work

Kazakh benchmarks. KazMMLU (Togmanov et al., 2025) is the largest Kazakh benchmark: 23,000 multiple-choice questions drawn from educational materials across STEM, humanities and social sciences, in both Kazakh and Russian. KazQAD (Yeshpanov et al., 2024) provides open-domain question answering with roughly 6,000 questions and passage-level relevance judgements. Qorgau (Goloburda et al., 2025) evaluates safety in the Kazakh-Russian bilingual setting. All are task-level evaluations in which the model selects or retrieves rather than produces inflected forms; none targets morphological generation, and none reports results conditioned on the phonological properties of the target.

Morphological inflection and its evaluation. The SIGMORPHON-UniMorph shared tasks have treated inflection as a supervised problem for a decade; the 2023 edition covered 26 languages from 9 families with lemma-disjoint splits designed to test generalisation to unseen lemmas (Goldman et al., 2023). Our item format — lemma plus target features, inflected form as output — follows that tradition, but evaluates instruction-following models zero-shot rather than trained inflection systems.

Closest to the present work, Ismayilzada et al. (2025) evaluate instruction-tuned multilingual models on Turkish and Finnish, treating morphemes as compositional primitives and testing productivity and systematicity over nonce roots. They report sharp degradation as morphological complexity rises and a lack of systematicity relative to humans. Critically for our purposes, their Appendix A.1 reports an instruction-language experiment: prompting in Turkish or Finnish rather than English yields "a drop or no change" of roughly 1-5 points. We differ in three respects. The language is Kazakh. The instruction language is the manipulated variable rather than a robustness check, and we decompose its effect by morphological category. And we evaluate zero-shot, which Section 5.3 shows is why our effect is an order of magnitude larger than theirs.

Prompt language and code-switching. Wendler et al. (2024) show that multilingual transformers process non-English input partly through English-centric internal representations, motivating the common practice of prompting in English regardless of target language. Zhang et al. (2023) find that multilingual models handle code-switched input poorly, which is directly relevant here: instructing in Russian or English while requiring Kazakh output is itself a code-switched configuration. Our design isolates the instruction-language effect on a task whose output is fixed in the low-resource language, so differences cannot be attributed to output-language fluency.

Finally, Arcon et al. (2026) evaluate metalinguistic knowledge across the 2,660 languages of the World Atlas of Language Structures, converting WALS features into multiple-choice questions, and find such knowledge to be fragmented and predicted mainly by digital resource availability rather than by genealogical or geographic factors. That converges with our Section 5.4 result at the level of a single

language, though it measures knowledge *about* languages rather than the ability to act on grammatical instructions *in* one.

3. The benchmark

3.1 Item construction

Each item pairs a citation-form Kazakh noun with a target morphological category and requires the inflected form as output. Eight categories are covered: plural, six cases (genitive, dative, accusative, locative, ablative, instrumental) and third-person possessive.

Gold answers are computed, not annotated. Suffix selection in Kazakh is determined by two independent factors. Vowel harmony assigns the suffix vowel according to the front/back quality of the last full vowel of the stem, so ұй takes -лер while тая takes -лар. Consonant assimilation selects the suffix onset according to the stem-final segment, partitioning consonants into five classes: vowels and *п/у/й* take *л*-initial plural suffixes; *л, м, н, ң, ж, з* take *д*-initial; voiceless segments take *т*-initial. Distinct partitions apply to each case. A third rule voices stem-final *п, қ, к* before vowel-initial suffixes, so *кітап* becomes *кітабы* and *сабақ* becomes *сабағы* under third-person possessive.

Encoding these rules as tables makes the item set reproducible from a random seed, free to extend to new stems, and free of annotator disagreement. It also permits conditioning the analysis on the properties that determine the answer, so that results can be reported by harmony class, by stem-final sound class, and by individual allomorph rather than only by category.

3.2 Lexicon and exclusions

Fifty regular noun stems are used, thirty back-harmonic and twenty front-harmonic, spanning all six stem-final sound classes. Stems that undergo syncope under suffixation — *халық* to *халқы*, *орын* to *орны*, *ауыз* to *аузы* — are deliberately excluded, since their surface forms are not derivable from the rules above. The benchmark therefore slightly understates the difficulty of real Kazakh.

Fifty stems and eight categories yield 400 unique items, from which a run samples 300 without replacement.

3.3 Validation

The rule tables are checked against 54 hand-verified forms before every generation; the generator refuses to emit items if any check fails. This guards against the failure mode in which a systematic error in the tables silently produces hundreds of incorrect gold answers.

3.4 Grading

Responses are graded by substring match against an accept list after removal of words echoed from the question. Echo removal is necessary because a model that merely repeats the prompt would otherwise score correct whenever the prompt contains a substring of the answer. Blank completions are treated as non-answers and excluded from the denominator rather than scored as errors (Section 6.4).

4. Experimental setup

4.1 Conditions

The instruction-language manipulation rewrites only the question field of each item. Identifiers, gold answers and accept lists are preserved byte-for-byte, so conditions are paired item-for-item and any difference is attributable to the instruction alone. In all conditions the required output is Kazakh.

Kazakh instructions use standard school grammatical terminology (табыс септігінде (кімді? нені?)). Russian and English instructions use the corresponding case names in those languages together with their conventional interrogative cues.

4.2 Models

Three models are evaluated: qwen2.5-3b and gemma3-4b served locally through Ollama, and gemma-4-31b-it accessed through a hosted API. gemini-3.6-flash was additionally used as a ceiling check on a subset. Decoding is greedy; repeated runs on identical items reproduce identical outputs.

Local models were given a 64-token budget, sufficient for single-word answers; API models required 1500 tokens for reasons discussed in Section 6.3.

4.3 Noise floor

Because items are sampled from a larger pool, run-to-run variation is non-zero. We estimate it by drawing three independent 300-item samples (seeds 7, 8, 9) and evaluating the same model under the same condition. All reported differences are compared against the observed range for the relevant category.

5. Results

5.1 Scale

Table 1. Accuracy by model and instruction language. gemma3-4b figures are means over three independent item samples.

Model	Params	Kazakh	Russian	English
qwen2.5-3b	3B	4.3	21.0	—
gemma3-4b	4B	16.6 $\pm$ 1.3	48.8 $\pm$ 1.0	52.0 $\pm$ 1.3
gemma-4-31b-it	31B	98.7	100.0	—

Others are single runs of 300 items (3B) and 150 items (31B).

Table 1 shows the same pattern in two small models: a large deficit under Kazakh instructions that a high-resource instruction language largely removes. At 31B the deficit is absent. Note that gemma3-4b and gemma-4-31b-it belong to different model generations, so the comparison confounds capacity with a generation of training improvements; a within-generation control at 12B was attempted but not completed for reasons of local disk (Section 8).

5.2 Instruction language by category

The overall difference between Russian and English is 3.2 points, marginally outside the 2.7-point noise band. Beneath that near-parity, individual categories diverge by an order of magnitude more, in both directions (Table 2):

Table 2. gemma3-4b accuracy by morphological category and instruction language. Mean over three item samples, plus or minus half the observed range.

Category	Kazakh	Russian	English	difference
plural	68.7 $\pm$ 6.0	58.9 $\pm$ 2.0	68.0 $\pm$ 1.5	EN +9.1
genitive	42.2 $\pm$ 1.5	1.8 $\pm$ 1.5	28.2 $\pm$ 3.2	EN +26.4
dative	1.7 $\pm$ 1.3	33.6 $\pm$ 6.8	48.4 $\pm$ 5.8	EN +14.8
accusative	0.0 $\pm$ 0.0	31.0 $\pm$ 5.4	62.9 $\pm$ 3.3	EN +31.9
locative	9.2 $\pm$ 1.5	76.7 $\pm$ 2.0	51.6 $\pm$ 2.4	RU +25.1

Category	Kazakh	Russian	English	difference
ablative	3.1 $\pm$ 0.5	57.9 $\pm$ 1.3	40.8 $\pm$ 5.9	RU +17.1
instrumental	11.0 $\pm$ 1.2	74.6 $\pm$ 4.5	53.4 $\pm$ 2.6	RU +21.2
possessive-3	2.6 $\pm$ 0.1	56.4 $\pm$ 1.4	62.4 $\pm$ 2.1	EN +6.0

Russian instructions are substantially better for the oblique cases — locative, instrumental, ablative — while English instructions are substantially better for accusative and genitive. Russian lexicalises an instrumental and a locative/prepositional case; English has no case morphology and its case terms are purely metalinguistic. Whether the interaction reflects that difference in grammatical inventory, or simply uneven quality across our own glosses, is not resolvable with the present design (Section 8).

5.3 In-context examples close the gap

The results above are zero-shot. Prior work on morphological evaluation in agglutinative languages prompts with in-context examples, and reports only a 1-5 point penalty for instructing in the target language rather than English (arXiv:2410.12656, Appendix A.1). We reconcile the two by varying the number of examples on our own items (Table 3). This experiment uses a single item sample (seed 7), whose zero-shot Kazakh baseline is 16.3 rather than the 16.6 mean of Table 1.

Table 3. Effect of in-context examples on gemma3-4b, by instruction language and number of demonstrations.

Category	KK 0-shot	KK 1-shot	KK 5-shot	RU 0-shot	RU 5-shot
plural	63.6	69.7	84.8	60.6	78.8
genitive	41.5	51.2	68.3	0.0	75.6
dative	2.4	70.7	80.5	41.5	63.4
accusative	0.0	22.5	67.5	25.0	72.5
locative	7.5	55.0	62.5	75.0	47.5
ablative	3.7	44.4	66.7	59.3	55.6
instrumental	12.5	70.0	87.5	75.0	85.0
possessive-3	2.6	52.6	78.9	57.9	81.6
OVERALL	16.3	54.7	74.7	48.3	70.3

A single example raises Kazakh-instructed accuracy from 16.3 to 54.7. Five examples raise it to 74.7, at which point it slightly exceeds the Russian condition: the 32.0-point instruction-language gap becomes -4.3. Kazakh gains 58.3 points from examples while Russian gains 22.0.

The gains are not uniform, and their distribution supports the account in Section 5.4. The categories that gain most from one example are exactly those where Kazakh instructions failed: dative +68.3, instrumental +57.5, possessive +50.0, locative +47.5, ablative +40.7. The two categories the model already handled under Kazakh instructions gain least: plural +6.1 and genitive +9.7. An example supplies what a grammatical term failed to convey — which transformation is intended — and it is redundant where the term was already understood.

The Russian genitive provides an internal check. Under our first Russian gloss it scored 0.0 (Section 6.1); with five examples it reaches 75.6. The model’s genitive knowledge was intact throughout; a mistranslated case label concealed it, and demonstrations bypass the label entirely.

Implication for evaluation practice. Few-shot prompting substitutes for instruction comprehension and therefore masks instruction-language effects. A benchmark that reports few-shot numbers is measuring something closer to analogical pattern completion than to a model’s ability to act on grammatical instructions in the target language. Both are worth measuring, but they are different

capabilities, and reporting only the former will understate how poorly a model serves users who write in that language.

Two caveats. With five same-category demonstrations the model observes most of the allomorph inventory for that category, so the 5-shot condition is closer to analogy than to rule application; the 1-shot figure is the more conservative one. And the Russian condition loses ground on locative (75.0 to 47.5) and ablative (59.3 to 55.6) under 5-shot prompting, which we do not currently explain.

5.4 The Kazakh penalty is category-specific

The deficit under Kazakh instructions is not uniform. For plural, Kazakh instructions perform as well as English (68.7 against 68.0). For genitive they outperform both alternatives (42.2 against 28.2 and 1.8). For accusative and dative they collapse to near zero.

The categories that survive Kazakh prompting are those whose Kazakh grammatical terms — көпше түрі, ілік септік — are basic school vocabulary; those that fail use more specialised terminology. This suggests that the bottleneck is coverage of Kazakh metalinguistic vocabulary rather than an inability to process Kazakh instructions in general, and predicts that term frequency in Kazakh corpora should predict which categories survive. We do not test that prediction here.

5.5 Neighbour-language interference

Under Kazakh instructions, both small models produce forms drawn from a better-resourced Turkic relative, and they draw on different ones. gemma3-4b produces Мектеpler for мектептер, grafting the Turkish plural -ler in Latin script onto a Cyrillic stem, and substitutes Latin ә for Cyrillic э. qwen2.5-3b produces Kyrgyz morphology (айтып берет where Kazakh requires айтып береді) and Kyrgyz lexical items.

This interference is largely absent under Russian and English instructions, which supports reading it as a symptom of failed instruction parsing rather than of a corrupted Kazakh lexicon.

5.6 Phonological conditioning

Where models are above the floor, back-harmonic stems outperform front-harmonic ones: 52.2 against 42.2 for gemma3-4b under Russian instructions, 25.5 against 13.8 for qwen2.5-3b. At 31B both classes reach 100%.

qwen2.5-3b scores 0/40 on instrumental under both Kazakh and Russian instructions — the -мен/-бөн/-пөн paradigm appears to be absent from the model rather than merely unreachable. gemma3-4b handles instrumental at 74.6 under Russian. Two models of comparable size therefore fail on different categories, and an intervention targeting either would need to be built against measured per-model gaps.

5.7 Error profile at ceiling

gemma-4-31b-it makes two errors in 300 items. One is lexical: ұл returns Олар ("they"). The other is the single genuine morphological error observed at this scale: сөз returns сөзнің where Kazakh requires сөздің, applying the post-nasal allomorph after a fricative. Vowel harmony is correct; assimilation is not. All other cuts of the data — both harmony classes, all six stem-final sound classes, every allomorph — are at 100%.

6. Methodological findings

We report five issues encountered in building this evaluation. Each cost substantial time to diagnose and each generalises beyond Kazakh.

6.1 A translated case label moved one category by 41 points. We first glossed Kazakh ілік септік as Russian родительный падеж (кого? чего?). Russian's кого?/чего? overlaps with the accusative and with negation constructions and does not signal possession; the model was pulled toward ablative and instrumental forms (шаң returning Шаңмен, түн returning Түннен). Genitive fell from 42.2 under Kazakh instructions to 1.8 under Russian. All six allomorphs scored near zero uniformly — the signature of a category never identified, as opposed to a phonological error, which would fail some allomorphs and spare others. A cross-lingual evaluation measures its own translations as much as it measures the model.

6.2 The Kazakh genitive resists isolated elicitation. Three attempts failed in three distinct ways. The кого? чего? gloss pulled toward oblique cases. A revised родительный падеж принадлежности (чей? чьё?) pulled toward the possessed form (шаң returning Оның шаңы, дос returning Досының). An izafet frame (___ сүпеті) scored zero at 4B but 93.9 at 31B, and its residual errors at 31B were confined to бет, бас and ic — exactly the nouns that form natural compounds.

The explanation is grammatical. Kazakh izafet Type II (бала кітабы) takes a bare possessor; Type III (баланың кітабы) requires a definite possessor that an isolated prompt does not supply. The frame is a Type II slot, so the bare stem is a legitimate answer and our genitive gold is wrong for those items. The construction that hosts the genitive does not require it. We therefore report the genitive with this caveat rather than treating its scores as a clean measure of morphological competence.

6.3 Hidden reasoning tokens consume the output budget silently. The hosted gemma-4-31b-it performs internal reasoning that counts against maxOutputTokens. At a 256-token budget, roughly 208 tokens went to reasoning and the answer was never emitted; the API returned an empty completion. Symptoms were 12-48% blank responses varying between runs and answers truncated mid-word. Requesting suppression of reasoning did not prevent it. Raising the budget to 1500 tokens eliminated blanks entirely and moved the measured score from 86.0 to 98.7. Any API-based evaluation of a reasoning-capable model needs a budget far above the visible answer length and an explicit blank-rate check.

6.4 Blank responses were being scored as wrong answers. Our harness initially treated only explicit API errors as errors, so empty completions counted against accuracy. A non-answer is not a wrong answer; it belongs outside the denominator, and it should be retried on resumption.

6.5 Generation-length truncation was ruled out rather than assumed. qwen2.5-3b produces long meta-commentary under Kazakh instructions and hit its 64-token ceiling. Re-running sixty items at 256 tokens produced exactly the same four correct answers while throughput fell from 0.6 to 0.4 items per second: the model generated more text and used it to elaborate rather than to answer. The low score is genuine.

7. Validity

gemini-3.6-flash answered the first seven Kazakh-instruction items correctly before quota exhaustion, including the two instrumental items on which qwen2.5-3b fails forty times out of forty. gemma-4-31b-it reached 98.7% with zero blank responses. The items are well-formed and answerable in Kazakh; low scores at small scale reflect model behaviour rather than a defective instrument.

8. Limitations

The benchmark covers fifty regular noun stems and eight suffix categories. It contains no verbs, no syntax, no irregular stems, and no measure of fluency or translation quality; a model could score perfectly here and still write poor Kazakh. Syncopating stems are excluded, so the set understates real difficulty.

Genitive results carry the caveat in 6.2. Plural differences between instruction languages fall inside the noise floor and support no claim.

The category-by-instruction-language interaction in 5.2 admits two explanations we cannot separate: the instruction language's own grammatical inventory, or uneven quality across our glosses. Given 6.1, the second is a live possibility. Distinguishing them requires multiple independent glosses per category per language, which we leave to future work.

The scaling comparison spans two model generations. Error bars are available for gemma3-4b only; other models were run once. The noise floor is estimated from item sampling and does not capture decoding variance, which is nil under greedy decoding, or variance across prompt phrasings, which 6.1 suggests is large. Three models establish a trend, not a scaling law.

9. Conclusion

Small multilingual models know more Kazakh nominal morphology than they can be asked to demonstrate in Kazakh. Instructing gemma3-4b in Russian or English rather than Kazakh raises accuracy from 16.6 to roughly 50 on identical items with identical Kazakh answers, and the deficit disappears entirely at 31B parameters. The gap is not uniform across the paradigm: it is largest for categories whose Kazakh grammatical terminology is least common, which locates the bottleneck in metalinguistic vocabulary rather than in morphological knowledge or in Kazakh processing as such.

For practitioners building Kazakh systems on small open-weight models, this suggests that instruction tuning on Kazakh task descriptions is likely to recover more capability per unit of compute than continued pretraining on Kazakh text. For evaluation, it suggests that any multiple-choice benchmark reporting a Kazakh-Russian gap is measuring a mixture of knowledge and instruction comprehension, and cannot separate them by design.

Availability

Code and item sets: <https://github.com/li-kun-g/kaz-eval> Archived release: <https://doi.org/10.5281/zenodo.21921339>

All items are generated from a seed; the generator validates its rule tables against hand-verified forms before emitting anything.

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